

■ Music Career Simulator

Development Reports

Internal changelog & feature specification · April 2025

v1.1

Health & Fatigue Fixes

9 April · 11:08 PM

Song Quality Behaviour

- Quality depletion begins only after fatigue exceeds **75%** — quality never reaches 0 until fatigue hits 100%.
- Once fatigue crosses 100%, quality slowly decays to 0 over time rather than dropping instantly.

Starting Values

- Chosen genre starts at **60**; all other genres default to **30**.
- Chosen skill starts at **60**; all other skills default to **25**.

Song Creation Quality Range

- With all factors (fatigue, health, skills, genre) producing a base score x , the range is always $(x - 2.5, x)$.
- While $\text{max quality} \leq 5$: score is drawn uniformly at random within that range.
- Once max quality exceeds 5, probability distribution becomes:
 - 20% chance $\rightarrow (x - 2.5, x - 2)$ — shitty track territory
 - 60% chance $\rightarrow (x - 2, x - 1)$ — average output
 - 10% chance $\rightarrow (x - 1, x - 0.5)$ — solid track
 - 10% chance $\rightarrow (x - 0.5, x)$ — peak quality

→ Ensures realism: even skilled artists occasionally produce below-average work.

Health Depletion

- Health depletion has been **reduced** — previous rate was too aggressive.

Fatigue & Skill Progression

- Skill additions increase fatigue cost as skill value rises. Threshold to trigger a skill gain in the **90–100** range requires the artist to be below **20% fatigue**.
- Extrapolated fatigue requirements across all skill ranges (proportional scaling from the 90–100 anchor):
 - 0–30 range $\rightarrow$ no fatigue restriction
 - 30–50 range $\rightarrow$ below 70% fatigue
 - 50–70 range $\rightarrow$ below 55% fatigue
 - 70–80 range $\rightarrow$ below 40% fatigue
 - 80–90 range $\rightarrow$ below 30% fatigue
 - 90–100 range $\rightarrow$ below 20% fatigue

Fatigue Recovery & Other Adjustments

- Fatigue recovery speed increased — artists should be able to bounce back more easily.
- Active course enrolment adds an extra **+15% fatigue** per week for the full **12-week** duration.
- Song creation now causes **less fatigue** than before.

Live Module — Sentence Pool Expansion

- All live-module sentence pools have been expanded with additional varied sentences.
- Long sentences are no longer repeated cyclically.
- Comments per cycle increased from ~10 to **15–20**.
- Repeated live sessions with low-viewer counts will reflect reduced comment diversity accordingly.

v1.2

Money & Popularity System

10 April · 4:22 PM

Popularity Meter

- Introduced a **Popularity** meter that starts at a minimal value and grows as the artist releases songs and goes live.
- New modules: **Management** and **PR**.
- Going live grants a **+5% base popularity** boost per session.

Currency & Side Hustle Jobs

- Currency is denominated in **USD (\$)**.
- Side hustle jobs introduced — examples below (names changed to avoid direct branding):
 - Fast-food crew member — **\$20 / week**, low fatigue cost
 - Additional jobs scale: more pay = more fatigue per week
- Genre courses now have an **in-game monetary cost** attached.

Media Management & PR Contracts

- Artists can sign contracts with one of **5 management companies**. Contract durations: monthly, 6-month, or yearly.
- Base price is negotiable around each company's listed rate.

Company	Contract Cost	Popularity Boost	Tier
Amplify Collective	\$1,000 / year	10 – 20%	Entry
The Narrative Arc	\$3,000 / year	20 – 25%	Standard
Momentum Media	\$10,000 / year	40%	Professional
Catalyst Comms	\$40,000 / year	60%	Elite
Verve Public Relations	\$100,000 / year	75%	Prestige

→ Management provides a weekly base popularity floor. Going live stacks an additional +5% on top of this.

Song Releases & Popularity

- Releasing a song provides an instant **+5% base popularity** boost in that week.
- In the following week, the released song contributes popularity proportional to its quality rating (e.g. a 10/10 adds 10%, a 5/10 adds 5%).
- Releasing an **album** grants a one-time **+15% popularity** boost in the release week only.

Stream Count & Song Quality

- Streams are driven by both **popularity (70%)** and **song quality (30%)**.
- A low-quality song on high popularity will still underperform; a high-quality song on low popularity will still do respectably.
- New module: **Shawtify Streams** — displays stream counts for all released songs.

Stream Earnings Formula

- Base rate: **\$0.01 per stream**. Milestone bonuses stack as follows:

Milestone	Bonus Earnings	Running Total
Base (per stream)	\$0.01 / stream	—
Every 10k streams	+\$10	Cumulative
Every 100k streams	+\$500	Cumulative
1M streams	+\$3,000	~\$19,000 total

→ Example: 1M streams = $(1M \times \$0.01) + \$3,000 + (100 \times \$10) + (10 \times \$500) = \$19,000$

- If signed to management, the management company deducts a negotiated **revenue-share** % from weekly stream earnings.
- Weekly earnings summary is displayed after each simulation turn, including the management cut breakdown.

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Job & Popularity Fixes

10 April · 5:27 PM

Side Hustle Job Rework

- Jobs now lock the artist into a **12-week commitment** — you cannot switch or stack jobs during an active contract.
- Pay is disbursed **once per week**; the weekly-choice-and-multi-activation exploit has been removed.
- Fatigue cost per job has been increased across the board to better reflect the workload.

Management Base Popularity — Corrected Behaviour

- Base popularity from a management contract is now a **fixed weekly value** — it does not stack or accumulate over time.
- Example: if a company provides 20% popularity, the artist sits at 20% from that source every week of the contract.
- When the contract expires, base popularity **gradually fades** to 0 over several weeks rather than cutting off instantly.
- Weekly base popularity varies within the company's stated range for realism (e.g. 17.5% one week, 21% the next).

Song-Driven Weekly Popularity Gains

- Each song's stream count this week contributes popularity according to the table below. Each tier has an individual cap, and total song-popularity gain is capped at **20% per week**.

Stream Threshold	Popularity Gain	Weekly Cap
1,000+	+0.5%	2%
10,000+	+1%	4%

100,000+	+3%	8%
1,000,000+	+5%	10%
Total song popularity cap	20% / week	—

Worked Example

Song	Streams This Week	Popularity Gained
Song 1	3,000	+0.5%
Song 2	15,000	+1.0%
Song 3	130,000	+3.0%
Song 4	7,000,000	+5.0%
Song 5	3,000,000	+5.0%
Song 6	14,000,000	+0.0% (cap reached)
Total	—	14.5% (below 20% cap)

→ Total song popularity gain = 14.5%. Had it exceeded 20%, it would be capped at 20%.

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Fatigue, Recovery, Management & Song Creation

11 April · 3:16 PM

Fatigue Recovery — Revised Formula

- Base recovery is now a flat **50 points per week**: if last week ended at 0% fatigue, next week starts at 50%.
- Recovery when the artist was NOT at 0% fatigue last week is penalised by any recurring fatigue cost:
→ **Formula: Recovery = Base Recovery – (x% × Base Recovery) = 50 – (x × 50)**
- where x is the fatigue cost ratio of the active recurring action (e.g. a job costing 25% fatigue → x = 0.25).

Scenario	Last Week Fatigue	Job Cost	Next Week Starts At
Case 1 — fresh start	0%	24%	0% + 24% deducted directly = 24%
Case 2 — ongoing fatigue	100%	25%	$100 - (50 - 0.25 \times 50) = 100 - 37.5 = 62.5\%$
Case 2b — moderate fatigue	52%	24%	$52 - (50 - 0.24 \times 50) = 52 - 38 = 14\%$

→ This prevents the exploit where a rested player's fatigue immediately jumps up from recovery then gets slashed by the job cost every cycle.

Management Contract Pricing & Plan Rework

- All management company fees have been **substantially increased**; base popularity values have been slightly reduced.

- Three plan tiers now exist with different billing cadences:
 - **4-Week Plan** — billed every week (weekly charge applies each of the 4 weeks)
 - **6-Month Plan** — billed every 4 weeks (pay the 4-week rate once per 4-week period)
 - **Yearly Plan** — billed in two instalments: 6-month cost upfront, then again at week 24

Negotiation System — Conversational Chat Model

- The negotiation UI is now a **full back-and-forth conversation chat** rather than a single prompt.
- The company presents their initial ask (fee + revenue cut) at the top of the chat.
- The artist can propose a counter-offer adjusting **fee and/or revenue-cut %**; the company replies in character.
- Company reaction tiers based on the counter-offer:
 - Significantly below ask → **furious response** — company may walk away
 - Slightly below ask → mild pushback, possible compromise counter
 - At or near ask → standard acceptance
 - Above ask → **grateful public response** + internal monologue calling the artist a fool

Genre Course Costs — Updated Pricing

Proficiency Range	Course Cost
10 – 40	\$250
40 – 60	\$1,000
60 – 80	\$5,000
80 – 90	\$20,000
90 – 100	\$100,000

Song & Album Management — New Permissions

- **Delete songs** — songs kept as singles can now be deleted.
- **Album-released songs on Instagram Live** — songs released as part of an album now also appear in the released-songs section during live streams.
- **Add singles to an existing album** — songs previously kept as singles can be added to an already-created album.
- **Add released songs to an album** — fully released songs can be retroactively added to an album.

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Fine-Tuning Popularity & Fatigue

12 April · 4:14 PM

Going Live — Popularity Rework

- Each live session now grants a flat **+2 popularity**.
- Weekly cap from going live: **+6 popularity** (i.e. max 3 lives count per week).
- Live-sourced popularity **decays the following week** to 1–2 points rather than carrying over in full.

Song-Driven Popularity — Weekly Reset Model

- Song popularity contributions are now **recalculated fresh every week** based on that week's stream counts only — nothing carries over.

- Within a tier, the gain is flat regardless of exact stream count (e.g. a song doing 1k and one doing 9k both add **+0.5** — it crossed the 1k threshold, not the 10k one).
- Individual tier caps and the overall 20% weekly cap remain in effect.

Worked Example — Week-by-Week Song Popularity

Week	Song Streams	Popularity Gained	Carries to Next Week?
Week 7	Song A: 3,000 Song B: 7,000	0.5 + 0.5 = +1.0	No — resets to 0
Week 8	Song A: 1,000 Song B: 700	0.5 + 0.0 = +0.5	No — resets to 0

→ Song B hitting 700 in week 8 contributes nothing — it didn't cross the 1k threshold that week.

Job Fatigue Cases — Revised Logic & Pay Penalty

- Let x = the fatigue cost % of the job. Three cases now govern how the week starts and how much the artist is paid:

Case	Last Week Fatigue	Next Week Starts At	Pay Received
Case 1	0% (fully rested)	$x\%$ fatigue applied directly	Full pay (m)
Case 2	Exactly $x\%$ (job cost only)	Starts at $x\%$ again	Full pay (m)
Case 3	More than $x\%$ (extra fatigue)	$50 - (x \times 50)$ recovery applied eg. 50% fatigue + 20% job → starts at 10%	$m - (y/2)\% \times m$ where y = extra fatigue % above x

Pay Penalty Formula (Case 3)

→ $\text{Pay} = m - (y / 2)\% \times m$ where m = full pay, y = extra fatigue % above job cost

Job	Last Week Fatigue	Extra Fatigue (y)	Calculation	Pay Out
\$100 job, 20% cost	50%	30% above 20%	$100 - (0.30/2) \times 100 = 100 - 15$	\$85
\$100 job, 20% cost	20% (= x)	0%	No penalty	\$100

→ Player-facing message (Case 3): "You were extra tired last week — your boss docked some of your pay."

→ Sets up future mechanics: repeated Case 3 occurrences could eventually trigger a lay-off.

■ Happy 9 Months, Teena ji & Me! ■

12th April 2025 · Nine months of us · logged forever in a music sim dev doc, as it should be



Release Popularity Bump — Quality-Tiered, Week 1 Only

- The release bump now applies **only in the first week** of a song's release and is scaled to song quality.

Song Quality Range	Popularity Bump	Duration
0.0 – 3.0	+1	Week 1 only
3.1 – 5.0	+2	Week 1 only
5.1 – 7.0	+3	Week 1 only
7.1 – 9.0	+4	Week 1 only
9.1 – 10.0	+5	Week 1 only

→ From week 2 onwards the bump is fully removed — no carry-over.

Popularity Decay — Faster Post-Contract Drop

- Previous decay was far too slow (15%% → 5%% taking 10 weeks). Post-contract popularity now falls **5–7%% per week**, with the exact value randomised each week for realism.
- The randomised range means no two contract expirations feel identical.

Stream Generation — Dynamic Range (No More Static Output)

- The stream formula now produces a **computed base value** which is then spread across a dynamic range — the same inputs will never produce the exact same number twice.
- The spread range scales with the computed stream value — larger songs have wider variance:

Computed Streams	Output Range (approx.)	Variance Width
5,000	3,000 – 6,000	~60%% below / 20%% above
150,000	110,000 – 160,000	~27%% below / 7%% above
1,000,000	Proportionally wider	Range scales with base

Song Stream Decay — 4-Week Grace, 12-Week Floor

- Songs do not decay for the **first 4 weeks** after release — they draw from the full dynamic range.
- From week 5 onward, a decay variable activates and gradually reduces streams until they hit **5%% of the original computed base** by week 12.
- After week 12, the song settles into a permanent low range based on 5%% of its computed base — also randomised within a tight band.

Week Range	Stream Behaviour	Example (base: 5,000)
Weeks 1–4	Full dynamic range, no decay	3,000 – 6,000 / week
Weeks 5–12	Decay variable active, declining	Gradually falling toward 250
Week 12+	Settled at 5%% of base, with range	150 – 280 / week (permanently)

→ Further decay stages (e.g. after 6 months) are planned for a future update.

Popularity-Linked Decay Floor — Old Songs Still Benefit

- The 5%% decay floor is **not fixed to launch-week numbers** — it is recalculated each week against the current popularity-adjusted stream formula.
- This means a song released 12 weeks ago on low popularity can still generate strong numbers if the artist has grown significantly since release.

Scenario	Launch Week	Week 12 Popularity	Week 12 Streams
Low → High popularity arc	5%% pop → 5,000 base streams	80%% popularity	5%% of 50,000 = 2,500/wk
Flat popularity	5%% pop → 5,000 base streams	5%% popularity	5%% of 5,000 = 250/wk

→ An artist blowing up at week 12 breathes new life into their back catalogue — old songs get rediscovered as new fans explore their discography. Realistic and rewarding.

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Introducing the Artist Ecosystem

13 April · 11:21 PM

Overview

- A pool of **50 pre-built artists** now exists in the world alongside the player. They release music, build popularity, react to the player, and interact with each other independently every week.
- Every ecosystem artist simulates alongside the player — the world keeps moving whether or not the player does anything.

Per-Artist Attributes

Attribute	Type	Description
name	string	Artist's display name
genre_proficiencies	dict {genre: int}	Skill level per genre (0–100), drives release quality
theme_proficiency	dict {theme: int}	Strength per theme — affects collaboration quality bonus
skills	dict (4 keys)	lyrics, vocals, production, mix/master — mirrors player skills
popularity	float 0–100	Current popularity in the ecosystem
release_tendency	float	How frequently this artist drops music (prob per week)
quality_consistency	float 0–1	How close to their ceiling they perform per release
friendliness	int 0–100	Governs personality type (friendly / neutral / rude) and ease of building relationship with player
controversy_meter	float 0–100	Higher = more likely to cause drama, appear in news, take jabs at the player or rivals

brutality	int 0–100	Response to disrespect: 0 = lets it go, 50 = disses back, 100 = escalates aggressively
first_release_week	int	The in-game week on which this artist drops their debut — staggers ecosystem so not everyone drops week 1
feature_cost	float (\$)	Price the player pays to feature this artist on a track

quality_consistency vs release_tendency — Now Separate Variables

- **release_tendency**: how often an artist drops music — a high value means they drop frequently, a low value means they go quiet for stretches.
- **quality_consistency**: how close to their ceiling they perform when they do release — an artist can drop rarely but always go 9/10, or drop every week but fluctuate wildly.
- These being separate creates realistic archetypes: the prolific-but-inconsistent artist, the slow-burning perfectionist, the hype machine who drops constantly but mid.

Theme Proficiency & Collaboration Bonus

- Each ecosystem artist has proficiency values per theme (heartbreak, street life, protest, etc.).
- When the player features an ecosystem artist on a track, the collaboration adds a quality bonus proportional to how well the featured artist's theme proficiency aligns with the song's theme.
- Example: Kendrick has high street life proficiency — featuring him on a street life track boosts quality more than featuring him on a party track.

Weekly Ecosystem Output — Release Feed

- After every week simulation, the game prints a release feed showing how many ecosystem artists dropped music that week, followed by individual release lines.
- Release line format:
→ **[Artist Name] releases [album / EP / single / mixtape] "[Generated Title]" — [Genre] / [Theme] — [Score]/10**
- The release title is fetched from the name generation file.
- The score is generated from the artist's skills, genre proficiency, and quality_consistency — integer only for ecosystem artists.

Staggered Debut System

- Each artist has a **first_release_week** variable — an in-game week number before which they will not release anything.
- This is randomised across the 50 artists at game start, spread across roughly the first 12–16 weeks, so the ecosystem builds organically rather than exploding all at once on week 1.
- After their debut, each artist then releases according to their release_tendency probability each week.
- The simulation logic mirrors the player's week loop — ecosystem artists tick forward on every simulate_week call, making the world feel synchronised.

Personality Types (derived from friendliness)

friendliness range	Personality Type	Behaviour
70–100	Friendly	Easy to build relationship with, less likely to jab, may reach out first
35–69	Neutral	Default state, reacts to what the player does, no strong lean

0–34	Rude	Hard to win over, higher base jab probability, low controversy threshold
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→ Reputation system (planned next) will expand artist interactions — controversy_meter and brutality are defined now but will fully activate once reputation is live.

Release Cooldown Periods — Format-Specific, Artist-Specific

- After releasing any project, an artist enters a cooldown period for that **format only**. Singles, EPs, mixtapes, and albums each have separate cooldown timers.
- Once the cooldown expires, the artist resumes releasing according to their release_tendency. Cooldowns are based on real release gaps derived from each artist's actual discography.

Artist	Single	EP / Mixtape	Album	Real-world basis
Frank Ocean	26 wks	80 wks	210 wks	Channel Orange (Jul '12) → Blonde (Aug '16) = ~212 wks. Singles sporadic since 2017.
Tyler, The Creator	12 wks	60 wks	120 wks	Flower Boy ('17) → Igor ('19) = ~104 wks; Igor → Call Me If You Get Lost ('21) = ~104 wks. Rare singles.
Kanye West	6 wks	30 wks	100 wks	MBDTF ('10) → Yeezus ('13) = ~130 wks; TLOP ('16) → ye ('18) = ~120 wks. Erratic singles.
Drake	2 wks	20 wks	52 wks	Albums ~every 12–18 months. Singles/loosies near-constantly. Views ('16) → Scorpion ('18) = ~110 wks but typically shorter.
J. Cole	10 wks	52 wks	96 wks	Born Sinner ('13) → FHD ('14) = ~78 wks; FHD → 4YEO ('16) = ~104 wks; KOD ('18) → The Off-Season ('21) = ~156 wks avg ~100.
Kendrick Lamar	8 wks	50 wks	130 wks	GKMC ('12)→TPAB ('15)=~130 wks; TPAB→DAMN ('17)=~104 wks; DAMN→Mr.Morale('22)=~260 wks; avg ~130.
Travis Scott	4 wks	35 wks	110 wks	Rodeo ('15)→Birds('16)=~52 wks; Birds→Astroworld('18)=~104 wks; Astroworld→Utopia('23)=~260 wks; avg ~140, skew lower.
21 Savage	4 wks	30 wks	70 wks	Issa ('17)→I Am>('18)=~52 wks; Savage Mode II ('20)→Her Loss('22, collab)=~110 wks. Active single dropper.
Future	2 wks	16 wks	52 wks	One of the most prolific in the ecosystem — drops albums near-annually, mixtapes/loosies constantly.
Lil Uzi Vert	3 wks	24 wks	80 wks	Luv is Rage ('15)→LiR 2('17)=~104 wks; Eternal Atake delayed significantly (~130 wks). Active single presence.
Playboi Carti	8 wks	50 wks	150 wks	Die Lit ('18) → Whole Lotta Red ('20) = ~130 wks; WLR → Music ('25) = ~220 wks. Notoriously slow dropper.
SZA	8 wks	50 wks	130 wks	CTRL ('17) → SOS (Dec '22) = ~286 wks. Very slow album cadence, sporadic singles.

Doja Cat	4 wks	30 wks	65 wks	Hot Pink ('19)→Planet Her('21)=~90 wks; Planet Her→Scarlet('23)=~115 wks. Active single presence.
Nicki Minaj	3 wks	30 wks	90 wks	Albums every ~2 years historically; Pink Friday 2 arrived after ~4 yr gap. Constant singles.
A\$AP Rocky	8 wks	40 wks	140 wks	ALLA ('15) → Testing ('18) = ~156 wks; Testing → Don't Be Dumb ('24) = ~320 wks. Infrequent dropper.
Metro Boomin	3 wks	20 wks	80 wks	Not All Heroes Wear Capes ('18)→Heroes & Villains('22)=~208 wks. Frequent producer features.
Juice WRLD	2 wks	16 wks	52 wks	Extremely prolific during his career — multiple loosies per month, albums ~annually.
Pop Smoke	2 wks	14 wks	40 wks	Very short career arc — Meet the Woo → Shoot for the Stars in ~26 wks. High frequency modelled.
Central Cee	3 wks	20 wks	65 wks	Wild West ('21) → 23 ('23) = ~100 wks. Active UK drill presence with consistent single drops.
Dave	8 wks	40 wks	110 wks	Psychodrama ('19)→We're All Alone in This Together('21)=~117 wks. Deliberate, infrequent dropper.

→ Cooldown values are per-format and independent — an artist can be in album cooldown while still dropping singles freely.
→ All cooldowns are the **minimum** floor. release_tendency still governs whether a release actually fires once cooldown expires — so a low-tendency artist may wait several more weeks beyond cooldown before actually dropping.

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Artist Release Logic & Name Generator

17 April · 12:30 PM

First-Release Guarantee Per Format

- Every ecosystem artist is guaranteed to have a **first release of each format** (single, EP, mixtape, album) at some point in their career — not all at once, but eventually, in an order that makes narrative sense.
- The stagger window for first releases has been extended from 12 weeks to **2 years (~104 weeks)**. Some artists drop their debut single on week 4, others won't surface until week 78 or later — making the ecosystem feel like a world that's been running, not a starting gun.

Relative Cooldown System — Cross-Format Logic

- Individual per-format cooldowns (from 1.3.1) remain intact. On top of those, a **relative cooldown** now prevents logically inconsistent back-to-back releases across formats.

Trigger Release	Blocks	Relative Cooldown	Notes
Album	Mixtape, EP	~52 weeks (artist-varied)	No mixtape or EP for ~1 year post-album. Makes narrative sense — the album IS the statement.
Album	Single	8–16 weeks (artist-varied)	At least a few months of silence after a major drop before singles resume.

Mixtape / EP	Mixtape, EP	~26–40 weeks	No back-to-back tape/EP drops within half a year.
Mixtape / EP	Single	4–8 weeks	Short quiet period post-tape before singles return.
Single	Nothing blocked	—	Singles don't trigger relative cooldowns on other formats.

Single → Project Pipeline (75%% Rule)

- When an artist drops a single, there is a **75%% chance** they are in a pre-project window — meaning an album, EP, or mixtape will follow within a dynamically calculated number of weeks.
- The follow-up project window is weighted by the artist's release_tendency and their current cooldown state. A high-tendency artist follows up faster; a reclusive one may still wait months.
- The remaining 25%% are true one-off singles with no project attached for that cycle.

Major Release Cooldown Floor — 1 Year Minimum

- After any album, mixtape, or EP release, a **hard floor of ~52 weeks** applies before any major project (album/mixtape/EP) can be considered again, regardless of individual cooldown values.
- Once this floor expires, release_tendency governs when the next major project actually fires — a low-tendency artist may still wait another year beyond the floor.

Name Generator — Format & Genre/Theme Segregation

- The name generator is now split into **four format-specific pools**: singles, EPs, mixtapes, and albums each draw from distinct naming conventions.
- Within each pool, names are further weighted by the release's **primary genre and theme** — a hip hop street life album draws from grittier, location-based title patterns; an experimental existential EP draws from abstract, fragmented naming conventions.
- Additional names have been added across all pools to reduce repetition as the ecosystem accumulates releases over many simulated weeks.

Format	Naming Style	Genre/Theme Influence
Single	Short, punchy, evocative — 1–3 words	Leans into theme directly (e.g. heartbreak → "Dissolve", party → "All Night")
EP	Compact concept titles, often with a number or date	Genre-coded — drill EP names feel different from soul EP names
Mixtape	Street-coded, confident, often location or persona-based	Hip hop / trap names skew harder; pop / r&b softer
Album	Full concept titles — metaphorical, narrative, or declarative	Theme-weighted most strongly — existential albums read differently to euphoria albums

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Ecosystem Integration in Main Game

17 April · 1:25 PM

Base Reputation Added to All 50 Ecosystem Artists

- Every ecosystem artist now carries a **base reputation** value (integer, 0–100) reflecting their standing in the industry at game start.
- Reputation is separate from popularity — popularity is how many people know you, reputation is what the industry thinks of you. Both are displayed.

New Main Menu Option — View Artists

- A new option has been added to the main action menu: **View Artists**.
- Selecting it prints a directory of all 50 ecosystem artists showing **name, reputation, and popularity** as integers — a quick-glance leaderboard of the world the player is competing in.

Artist Profile View

- Selecting an individual artist from the directory opens their full profile. The profile displays:

Field	Display
Name	Artist display name
Reputation	Integer 0–100
Popularity	Integer 0–100
Friendliness	Integer 0–100 + derived personality label (Friendly / Neutral / Rude)
Skills	lyrics, vocals, production, mix/master — all as integers out of 100
Genres	All genre proficiency values above threshold, shown as integer scores
Themes	Theme proficiency values — highlights their strongest themes
See Releases	Sub-option to view full discography (see below)

See Releases — Discography View

- Choosing **See Releases** from an artist profile prints their full release history in a formatted table, sorted chronologically.
- Each entry shows four columns:

Title	Type	Review	Release Date
Lost Meridian	Album	8/10	Year 1, Week 14
Cold Circuit	Single	7/10	Year 1, Week 8
Pavement Gospels	Mixtape	6/10	Year 2, Week 31

→ *Release date uses in-game Year/Week format consistent with the main simulation clock.*

New Releases Feed — Post-Simulation

- After every week simulation, a **New Releases** section prints below the weekly summary showing every ecosystem artist who dropped music that week.
 - Uses the same ecosystem simulation logic already built — no new computation, just surfaced in the main game output.
 - Format: artist name, release type, title, genre/theme, score — identical to the ecosystem feed spec from 1.3.1.
 - If no artists released that week, the section is omitted silently.
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Overview

- The player now maintains a **relationship meter** (0–100) with every ecosystem artist. Each artist's **friendliness** and **lovingness** values (imported from the artist seed) govern how they respond to interactions and romantic advances respectively.
- Relatively friendly artists will interact with the player even without a prior relationship. Low-friendliness artists ignore or dismiss the player until reputation backs them up — that gate will be enforced once the reputation system lands.

Friendship Rating Tiers

Meter Range	Label	What it feels like
0 – 10	They strongly hate you	Active hostility — they may take jabs at you unprompted
10 – 20	Doesn't know you / wants to forget you	Cold dismissals, no engagement
20 – 30	Barely knows you	One-word replies, no warmth
30 – 40	More like a colleague	Professional but distant, occasional acknowledgement
40 – 50	He digs you	Warm enough to help with small things
50 – 60	You are friends	Genuine back-and-forth, relaxed tone
60 – 70	You are good friends	Goes out of their way for you sometimes
70 – 80	You are good good friends	Tells you things, shows up, has your back
80 – 90	Get a room you two	Industry notices the closeness, affects perception
90 – 100	Ride or die kinda shit	Full loyalty — would diss someone for you

Interaction Options

- The player can interact with any artist from the **Manage Relationships** menu. Multiple interactions can be done per session — the menu loops until the player opts out.
- Only **relationship gains are capped once per week** per artist — you can still interact more, but repeated praise/positivity won't keep increasing the meter. Negative interactions (criticism, spamming requests) still apply their full downward effect regardless of frequency.

Interaction	Effect on Meter	Notes
Text about their music (praise)	+1.5 to +15 (3× friendliness-scaled)	Gain capped once per week. Most reliable builder.
Give them a shoutout publicly	+3 to +12	Higher gain, slightly less frequent effect.
Ask for a gig slot	–2 to +5 (depends on tier)	Accepted warmly at 40+, awkward below 30, declined below 20.
Request money / studio help	–3 to –8 if declined; –1 if accepted	Depends on relationship tier AND artist wealth. Spamming tanks the meter.

Ask them out on a date	Depends on lovingness + relationship	See lovingness logic below. Can backfire badly if meter is low.
Criticize their recent interview	-5 to -20 (brutality-scaled)	High-brutality artists react much harder. Can trigger public jab.
Compliment their recent release	+2 to +9	Only works if they have a recent release (last 4 weeks).
Check in (casual message)	+0.5 to +3	Small but keeps decay from triggering. Good for maintenance.
Invite to collab	+2 to +8 if accepted	Feature cost applies. Relationship must be 40+ for consideration.
Apologize (after conflict)	+3 to +10	Only available if meter dropped in last 2 weeks. Can recover damage.
Talk about another artist	Variable	Praising their rival = -5. Dissing their rival = +3 to +8 if they have beef.

Relationship Gain Formula — 3× Multiplier

- All positive interaction gains are multiplied by **3×** to make friendship feel achievable without feeling instant.
→ ***Gain = base_gain × 3 × (friendliness / 100)** — so a 90-friendliness artist gives nearly full gain; a 10-friendliness artist gives almost nothing.*

Lovingness Variable — Date Interaction Logic

- lovingness** (0–100) is imported from the artist seed alongside friendliness. It governs receptiveness to romantic/date interactions independently of general friendliness.
- A high-lovingness artist who barely knows you might still entertain it. A low-lovingness artist with a great relationship will still shut it down politely. Both variables must align for a yes.

lovingness	Relationship Tier	Outcome
70+	40+ (digs you)	Yes — warm, genuine response
70+	Below 40	Soft maybe / not yet — door left open
40–69	60+ (friends)	Considered yes, slightly guarded
40–69	Below 60	Polite no — no relationship damage
Below 40	Any	No — tone depends on relationship tier
Any	Below 20	Dismissal — possible relationship damage

Response Voice — Friendliness-Differentiated Writing

- Sentence pools for responses are **distinct per friendliness tier** — the same rejection reads completely differently depending on who's saying it.

Scenario	High Friendliness (70+)	Low Friendliness (below 30)
Declining a money request	"I really can't right now, I'm sorry my brother."	"Nah. Who are you again?"

Declining a date	"You're sweet but the timing isn't right for me."	"Lol no."
Responding to praise	"Means a lot bro, genuinely. Thank you."	"Okay."
Ignoring an interaction	(reads message, no reply)	(leaves you on read immediately)

Relationship Decay & Damage

- No interaction for **3+ consecutive weeks** triggers passive decay — the meter drops 1–3 points per week until contact is made again.
- Criticism, spamming money/date requests, and praising their rival all apply immediate downward pressure. The steeper the drop, the harder it is to recover without an apology interaction.
- Decay does not apply if the meter is already below 20 — you can't fall further than cold.

Main Menu — Manage Relationships

- New main menu option: **Manage Relationships**.
- Selecting it shows all 50 artists with their current relationship meter value and friendship tier label at a glance.
- Choosing an artist opens their interaction screen, which also displays: relationship meter bar, friendship label, artist **lovingness** value, and their current week's mood (derived from recent interactions).
- The interaction menu loops — the player keeps choosing options until they select **Leave**. This allows natural multi-step conversations in a single session.

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Relationship-Aware Live Module

18 April · 12:38 PM

Artist List — Replaced with Ecosystem

- The hardcoded 20-artist sample list in the live module has been deleted. The **50 ecosystem artists** are now the source for all praise and criticism targets on live.

Relationship-Reflected Live Points

- **Praising** an artist on live gives relationship points in the range of **+2 to +8**, scaled by the artist's friendliness. This gain is capped once per week regardless of how many times you praise them on live.
- **Criticizing** an artist gives **–5 to –10** relationship points. There is no weekly cap on criticism — each instance applies its full damage.

Chat Sentence Pool Architecture — Layered Awareness

- The live chat now assembles its response from **multiple pools simultaneously**, stacking based on active conditions. The more context the game has, the richer and more unpredictable the chat output becomes.

Pool Type	Trigger Condition	Sentences
Normal	Always active — baseline reaction to praise/criticism	15–20
Relationship-aware	Triggered when player's relationship with the target is notable (good or bad)	15 per scenario (4 scenarios)

Memory	Triggered when player has a recorded history with the target (last 3 mentions)	15–20 per pattern (10 patterns)
Fatigue	Fatigue ≥ 85 — chat notices artist is tired	Existing pool (from live module)
Overlive	4th+ live this week — chat notices over-streaming	Existing pool (from live module)

Relationship-Aware Sentence Pools (4 scenarios, 15 sentences each)

Scenario	Trigger	Example lines
Good relationship, criticize	Relationship ≥ 50 , action = criticize	"what the hell??" / "i thought they were cool" / "and i thought i'd see them on a track next album" / "here goes another good industry friendship"
Bad relationship, criticize	Relationship < 10 , action = criticize	"yea i saw that coming" / "obviously dawg" / "finally" / "hell yeah this has been brewing for a long time now"
Bad relationship, praise	Relationship < 20 , action = praise	"is he kidding?" / "no way you meant that" / "stop the cap, we know you hate them"
Good relationship, praise	Relationship ≥ 60 , action = praise	Warm validation — "makes sense they're close" / "collab incoming fr"

Artist Mention Memory — Last 3 History Variable

- The game maintains a per-artist memory variable storing the player's last **3 mentions** of that artist across all lives: e.g. **[praise, praise, criticize]** (oldest $\rightarrow$ newest).
- Memory pools only activate from the **2nd mention onward**. First-time praise and first-time criticism use only normal + relationship-aware sentences.

Memory Pattern	Current Action	Chat Reaction	Example lines
[praise] $\rightarrow$ criticize	Criticize	Switching up	"why you switching up?" / "didn't you praise them a while ago" / "what happened all of a sudden?"
[criticize] $\rightarrow$ praise	Praise	Reversal surprise	"wait so you're cool now?" / "squashed that quick" / "that was fast work"
[praise, praise] $\rightarrow$ praise	Praise	Consistent love	"he always shows love" / "bro is literally their biggest fan"
[criticize, criticize] $\rightarrow$ criticize	Criticize	Ongoing beef	"this is a whole pattern now" / "bro won't let up" / "it's personal"
[praise, criticize] $\rightarrow$ any	Either	Confusion / drama	"nobody knows where they stand with this guy" / "chaotic energy every live"
[criticize, praise] $\rightarrow$ any	Either	Redemption arc / flip	"okay so they're moving on" / "beef is dead confirmed?" / "plot twist"
[praise, praise, criticize]	Criticize	Betrayal energy	"bro was defending them two lives ago" / "the fall from grace is real"
[criticize, criticize, praise]	Praise	Reconciliation	"the beef is over??" / "they talked it out fr" / "glow up for both of them"

[praise, criticize, praise]	Praise	Unpredictable	"i never know what to expect" / "these two are complicated" / "therapy arc"
[criticize, praise, criticize]	Criticize	Can't commit	"make it make sense" / "hot and cold every week" / "just drop a diss already"

→ 10 memory patterns × ~15–20 sentences each = 150–200 memory-specific lines in total.

Full Stack Example — How Pools Combine

- Pools are drawn from simultaneously. Each message in the chat has a weighted probability of coming from any active pool, keeping the output varied rather than cycling through one pool exhaustively.

Scenario	Active Pools
First time criticizing an artist	Normal + Relationship-aware (bad rel)
Praising someone you've criticized before	Normal + Relationship-aware + Memory [criticize → praise]
4th live, fatigued, criticizing an artist you have bad relationship with, and you've criticized them before	Normal + Relationship-aware (bad rel, criticize) + Memory [criticize, criticize] + Fatigue + Overlive
Good friends, praising them, first mention	Normal + Relationship-aware (good rel, praise)

→ The more context the game accumulates, the richer and more unpredictable each individual chat message becomes — the same action never generates the same feed twice.

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Introducing Features & Feature Requests

20 April · 12:42 AM

Overview

- A fully bidirectional feature system. Artists can request the player, the player can request artists. Both sides evaluate quality and relationship before accepting. Both sides can reject. Time limits apply. Everything has relationship consequences.

Inbound — Artist Requests the Player

- An ecosystem artist requests the player for a feature when **either** condition is true:
 - Player popularity is within range of the artist's popularity
 - Relationship meter with the player is > 90 — they want you regardless of clout
- Whether the request actually fires is gated by the artist's **release_tendency** — a low-tendency artist rarely asks even if conditions are met.
- The request arrives with a **song quality value** attached, generated from the artist's quality_consistency and skills in the relevant genre.
- Player can **accept or reject** — the choice is reflected in the relationship meter.

Outbound — Player Requests an Artist

- Flow: **Create a song** → **Manage Songs** → **Send Feature Request** → **Choose Artist**.
- Two pricing methods:
 - **Method 1 — Full cost.** Any artist is approachable. Pay their feature_cost upfront.
 - **Method 2 — Friendship rate.** Relationship > 90 → pay only 20%% of the base feature_cost.

Artist Acceptance Logic — Song Quality × Relationship

- When the player sends a feature request, the artist evaluates both the song quality and the relationship before responding. The following gates apply:

Relationship	Song Quality Requirement	Response Tone
> 90 (ride or die)	Accepts anything — even below 3 (80%% chance)	Warm, immediate yes
> 60 (good friends)	Song quality ≥ 5.0	"Sure buddy, I'll do it"
> 80 (close friend)	Song quality ≥ 5.0 (overrides consistency gate)	Accepts on relationship alone
10–30 (cold/colleague)	Threshold set by quality_consistency (see below)	Purely professional — song must earn it
Any	Song quality < 3.0	Hard rejection unless relationship > 90

Quality Consistency → Minimum Song Threshold (cold relationships only)

- For artists the player doesn't know well, the artist's own quality_consistency determines the minimum song quality they'll accept — high-consistency artists have high standards:

quality_consistency	Min. song quality required	Logic
> 0.90	9.0 / 10	Perfectionist — only elite songs earn their attention
0.80 – 0.90	7.5 / 10	High standard — strong songs only
0.70 – 0.80	6.5 / 10	Solid standard — above average gets a yes
< 0.70	5.0 / 10	Inconsistent themselves — can't be too picky

Working on Your Part — The Verse Loop

- Once a request is accepted (either direction), the player can access it via **View Feature Requests**.
- Selecting an active request opens the work loop: roll a verse quality based on the player's current skills and the song's genre, using the existing quality formula.
- Each roll costs **fatigue**. Player can roll as many times as they want, keeping the result they're happy with.
- When satisfied, player selects **Send this verse** — locking in the quality and advancing the request status.

Artist Delivers Their Part — Player Evaluates

- After accepting an outbound request, the artist takes their time to deliver — each artist has a **feature_turnaround** value (2–6 weeks) that reflects their working speed.
- When the verse arrives, the player sees the quality score and chooses to **accept or reject**.
- This is a genuine creative decision, not just a stat check:
 - Eminem sends a 6/10. You can accept because it's Eminem and his name alone elevates the song. Or you can reject because he's dragging down a 9/10 track.
 - Dijon sends a 9/10. Easy yes. But rejecting it — however unlikely — would end the collab and hurt the relationship badly.
- Rejection messages vary by artist personality — warm artists apologise, cold artists don't explain themselves.

4-Week Deadline — Time Kills Collabs

- The player has **4 weeks** from accepting a request to send their verse. If the deadline passes without a submission:
 - Opportunity is lost — the request expires
 - Relationship with the artist drops significantly
 - Player reputation takes a small hit — word gets around that you don't deliver
 - Player-facing message: *"You missed the deadline on [artist]'s feature. They've moved on."*

Quality Impact — How Features Affect the Final Song

- When a feature is successfully completed and added to a song, the song's quality is recalculated as a weighted blend:

$$\rightarrow \text{final_quality} = (\text{song_quality} \times 0.7) + (\text{feature_verse_quality} \times 0.3)$$

- A great verse raises the song. A weak verse from a big name is a real decision — the name adds streams but the quality metric drops.

Overview

- Projects (albums, EPs, mixtapes) now have real songs inside them, with names, genres, themes, qualities, and credits. The ecosystem simulation becomes fully musical — every release is now a tracklist, not just a title and score.

Project Song Counts

Format	Song Count Range	Notes
EP	3 – 6 tracks	Compact concept release
Mixtape	6 – 25 tracks (rarely beyond)	Street tape feel — longer, looser
Album	7 – 25 tracks (50+ theoretically possible)	Full statement — rarely exceeds 25 in practice

project_making_abilities — New Artist Attribute

- Added to the artist ecosystem seed. Governs **consistency** (how similar in quality the songs are across the project) and **cohesion** (how unified the genre/theme palette is).
- Genre and theme of individual songs vary slightly from the project's core genre/theme, with the variance inversely proportional to cohesion — a high-cohesion artist makes focused albums, a low-cohesion artist wanders.

Project Score Formula

- Final project score is computed from song qualities with four factors — reviewer bias excluded (that lives in the critics module, not here):

Factor	Description
Song quality average	Mean of all track scores — the foundation
Consistency bonus	Tighter quality variance across songs = higher bonus
Cohesion bonus	Genre/theme unity across the tracklist rewards focused projects
Boring penalty	Projects with no song above 7.5 get a small penalty — nothing stands out

Pre-Release Singles Integration

- Any single the artist released in the **last 8 weeks** before a project is automatically included in the tracklist.
- This mirrors real industry behaviour — singles are advance cuts from an album, not separate from it.

Project Feature Requests

- While building a project, an artist may send the player a feature request for **one of the songs in the album** — using the existing feature request system from 1.3.6.
- If the player's feature is accepted and completed, the song title reflects it:
→ *hunni ft. dijon* (the ft. credit appears in the song title within the project tracklist)

Post-Completion Release Delay

- After a project (or song) is finished, the artist doesn't release it immediately. There is always a delay of **2 to 12 weeks** between completion and release — realistic label timing, rollout strategy, and press cycles.

Artist Social Graph — Friends & Enemies

- Every artist now has a **friends list** and a **dislike list**. These are generated from the artist seed based on friendliness and controversy values:

Artist Type	Friends List Size	Dislike List Size	Basis
High friendliness (e.g. Clairo)	15–20 artists	2–3 artists	Warm, broadly liked
Mid friendliness (neutral)	8–12 artists	5–8 artists	Selective but social
Low friendliness (e.g. Eminem)	6–7 artists	10–15 artists	Few close, many distant
High controversy (>90)	Varies	On ~60%% of others' dislike lists	Controversial = widely disliked

- Artists never appear on both their own friends AND dislike list simultaneously.
- Artists will **not collaborate** with anyone on their dislike list.

→ *Friends list and enemies list are displayed in the View Artists profile screen.*

Collaboration Frequency by Friendliness

- How often an artist features others on their songs is directly tied to their friendliness:

Friendliness	Feature Rate	Character
High (70+)	~60%% of songs have a feature	Collaborative by nature
Mid (35–69)	~40%% of songs have a feature	Selective guest appearances
Low (<35)	~20%% of songs have a feature	Mostly solo — features are earned

Feature Selection — Genre & Theme Compatibility

- When choosing a collaborator, the artist primarily searches their friends list for artists whose **genre proficiency matches the song's genre**. A random selection is made from eligible artists.
- In 5%% of cases this breaks — cross-genre features happen (e.g. Eminem ft. Bon Iver on a hip hop heartbreak track — rare but possible, and interesting when it happens).
- Theme compatibility is also checked — for a heartbreak track, the artist strongly prefers collaborators with high heartbreak theme proficiency. Low-probability exceptions still exist.

Multiple Features — Quality Cap System

- Songs can have up to **5 features** (not counting production). Each feature pushes the quality up or down based on the featured artist's quality_consistency.
- The final song quality is **capped at ±2 from the original** quality before any features. This prevents infinite stacking:

Step	Action	Quality	Cap Check
Base	Eminem writes song	7.0	Base = 7.0, cap window = 5.0–9.0
+Kendrick	Strong verse, quality rises	8.2	Within cap ✓
+Drake	Good verse, quality rises	8.7	Within cap ✓

+Travis	Energetic verse, quality rises	9.2	Exceeds +2 cap → floor set at 8.8
+Kanye (weak)	Bad verse, quality falls	8.5	Negative movement allowed ✓

→ Upward movement beyond +2 is blocked. Downward movement is always allowed — a bad feature can always hurt a song.

- Song quality rarely exceeds **9.5**. A perfect 10 is possible but only at a **3–4%** chance.

Production Credits & Producer Classification

- Artists with **production skill > 70** are classified as producers and appear in other artists' production searches.
- Producers have genre affinities — a rock song won't be produced by Hit-Boy unless his genre profile supports it.

Production Skill	Behaviour	Quality Impact
> 70 (producer tier)	Self-produces 60–80%% of own songs. Appears in others' producer search.	Full quality — high production skill
45 – 70 (intermediate)	Self-produces 20–30%% of songs	–0.5 to –1.5 quality penalty on self-produced tracks
< 45 (non-producer)	Never self-produces — always seeks an external producer	Quality reflects chosen producer's skill

→ If no genre-matching producer is found in the ecosystem, the song defaults to self-produced regardless of skill level.

Song Title Format

- All song credits appear in the title string, following a consistent format:
 - *eminem ft. kendrick lamar (prod. Mike Dean)*
 - *kanye west - drugs ft. kendrick lamar (prod. kanye west) — self-produced*
 - *hunni ft. dijon (prod. kaytranada) — player featured*
- Multiple features appear comma-separated: **ft. kendrick lamar, drake, travis scott**

Enmity Logic — Revised Rules

- Enmity is now driven by three specific causes, not random assignment:
 - The artist themselves has a **high controversy rate** — they create friction by nature
 - Another artist has a **high controversy rate** — controversial people attract dislike
 - Another artist has **very low friendliness** — coldness breeds resentment over time
- Dijon example: low controversy, warm personality — his enemies only exist because *they* are too controversial for him to tolerate, not because he creates conflict.

Who Cannot Be an Enemy

- Artists with **high friendliness** (e.g. Dijon, Bon Iver) should rarely appear on anyone's enemies list — they're too universally warm to generate antagonism.
- Artists with **similar controversy levels** are unlikely to feud — a 70–80 controversy artist and an 80–90 controversy artist understand each other's chaos and largely coexist.
- A small random element remains — some feuds are inexplicable and that's realistic — but the total enemies list is **capped at 10** even for the most controversial artists.

Artist Profile	Max Enemies	Enemy Source
Low controversy + high friendliness (Dijon)	2–3	Only highly controversial artists who offend his sensibility
Mid controversy + mid friendliness	5–6	Mix of controversial and cold artists
High controversy (Kanye)	~20	Broad dislike — but capped at 10 going forward
High controversy (>90)	Hard cap: 10	On ~60%% of other artists' dislike lists passively

Artist Skill Categorisation

- Artists are now categorised by their **dominant skill set**, which governs feature eligibility, collaboration titles, and what role they play in a song's credits.

Category Formula

- Let H = the artist's highest skill value. An artist **possesses a skill** if it falls within the following range:
 - If $H > 50$: **skill qualifies if value $\geq H - 30\%(H)$ i.e. $\geq 0.7 \times H$**
 - If $H \leq 50$: **skill qualifies if value $\geq H - 20\%(H)$ i.e. $\geq 0.8 \times H$**
- If $H < 30$: the artist is a **Growing Artist** — they only collaborate with other growing artists and cannot appear in mainstream feature/production searches.

Artist (example skills)	Calculation	Categories
lyrics 20, vocals 5, prod 75, mix 67	$H=75$. $0.7 \times 75 = 52.5$. Prod 75 ✓, Mix 67 ✓, Lyrics 20 ✗, Vocals 5 ✗	Producer + Engineer
lyrics 80, vocals 95, prod 70, mix 12	$H=95$. $0.7 \times 95 = 66.5$. Vocals 95 ✓, Lyrics 80 ✓, Prod 70 ✓, Mix 12 ✗	Lyricist + Vocalist + Producer

lyrics 45, vocals 39, prod 21, mix 14	H=45. 0.8×45=36. Lyrics 45 ✓, Vocals 39 ✓, Prod 21 ✗, Mix 14 ✗	Lyricist + Vocalist
lyrics 28, vocals 25, prod 11, mix 5	H=28 (<30). Growing artist rule applies.	Growing Artist — restricted collabs only

Category Rules — What Each Category Can Do

Category	Can appear in ft. credit	Can appear in prod. credit	Releases solo music
Lyricist / Vocalist	Yes	No	Yes
Producer	No (unless also Lyricist/Vocalist)	Yes	Yes (if also L/V)
Engineer	No	No	No — sole engineers don't release
Growing Artist	Only with other growing artists	Only internally	Yes (own ecosystem)

→ Mike Dean (prod 75, mix 68, lyrics 15): Producer + Engineer only. Cannot appear in ft. credits. Appears in (prod. Mike Dean) in title.

→ Kaytranada (prod 88, vocals 60+): Producer + Vocalist. Can appear in both ft. and prod. credits.

Feature Request UI — Category-Filtered Search

- When browsing artists for a feature request, the player now sees artists grouped by category:
 - Check Lyricists
 - Check Vocalists
 - Check Producers (adds to prod. credit, not ft.)
 - Check Engineers (adds to engineered by metadata only)

Engineering Logic — Mirrors Production System

Mix/Master Skill	Behaviour	Quality Impact
> 70 (engineer tier)	Self-engineers 60–80%% of own songs. Appears in others' engineer search.	Full quality — skilled mixing
50 – 70 (intermediate)	Self-engineers some tracks (quality penalty applies)	–0.5 to –1.5 on self-engineered tracks
< 50 (non-engineer)	Always uses an external engineer available in that genre	Quality reflects chosen engineer's skill

- Engineers apply the same **±2 quality cap formula** as producers and featured artists.
- Engineers are **never mentioned in the song title** — their credit only appears in song metadata.

Song Metadata View

- The player can select any song from a released album (including ecosystem artist releases) and view its full metadata breakdown:

Field	Example
Title	arose ft. kendrick lamar (prod. Mike Dean)
Lyricists	Eminem, Kendrick Lamar

Vocalists	Eminem, Kendrick Lamar
Producers	Mike Dean
Engineered by	Mike Elizondo
Genre	Hip Hop / Heartbreak
Quality	9.1 / 10
Release week	Year 2, Week 14

- Artists who are only producers or engineers do **not** appear in Lyricists or Vocalists fields.
- Growing artists only appear in metadata for songs within their own bubble.

Title Format — Updated Rules

- *ft. only shows Lyricists / Vocalists — never producers or engineers alone*
- *(prod. xyz) shows in title — engineer credit is metadata only*
- *Self-produced: kanye west - drugs ft. kendrick lamar (prod. kanye west)*
- *External prod: eminem ft. kendrick lamar, drake (prod. Mike Dean)*

New Per-Song Variables

- Three new optional fields added to the shared **Song** model:

Variable	Range	Description
catchiness	0.1 – 5.0	How immediately hook-driven the song is. Applied as a piecewise multiplier on weekly streams.
virality	1 – 5	How likely the song is to spread beyond the core audience. Triggers bonus stream events post-maturity.
maturity_weeks	100 – 200 (testing)	How many weeks after release until the song hits its virality peak. Settles at ~30%% of peak thereafter.

- On **Create Song**, the game now rolls and prints all three values so the player can see what kind of song they've made before releasing.
- Songs created before this patch have catchiness/virality/maturity **auto-assigned at release time** so no existing saves break.

Catchiness & Virality Probability Buckets

Variable	Range	Probability	Notes
Catchiness	0.1 – 1.0	Most common	Average hook, background listen
Catchiness	1.0 – 2.0	Common	Noticeable hook
Catchiness	2.0 – 3.0	Uncommon	Earworm territory
Catchiness	3.0 – 4.0	Rare	Banger — people replay it
Catchiness	4.0 – 5.0	Very rare	Generational hook
Virality	1 – 2	Most common	Organic, word of mouth
Virality	2 – 3	Common	Picked up by playlists / blogs
Virality	3 – 4	Uncommon	Goes semi-viral, trend potential
Virality	4 – 5	4%% chance	Full viral moment
Virality	5	2%% chance	Once-in-a-career phenomenon

Revised Stream Formula

- Base multiplier reduced from **25,000,000** → **10,000,000**. The old value was producing ~9.7M streams for a 6.6 quality / 60 popularity song — far too high for a realistic early-career simulation.

→ **base_streams** = $10,000,000 \times (\text{weighted_score}^{1.85}) \times (\text{quality_factor}^{1.35})$

- Non-viral weekly base is capped at **150M**. Virality bonuses can push above this cap.

Step	What happens
1. Base curve	$10M \times \text{weighted_score}^{1.85} \times \text{quality_factor}^{1.35}$

2. Decay	Existing decay multiplier applied (4-week grace → 12-week floor)
3. Catchiness multiplier	Piecewise curve applied — higher catchiness = non-linear stream boost
4. Virality bonus	Activates at maturity_weeks. Peaks at week 8 post-maturity, decays to 30%% of peak by week 16, settles there permanently
5. Dynamic range	Final value spread across a range that scales with stream size (existing system)

- SHAWTIFY STREAMS now displays **catchy X** and — if virality ≥ 2.0 — **viral Y (matures in Zw)** alongside each song for testing visibility.

Ecosystem Artist Streams

- Ecosystem artists now generate streams for their released songs each week using the **same formula** as the player — same base curve, same decay, same catchiness/virality system.
- Catchiness and virality are randomly generated for ecosystem songs using the same probability buckets as the player.
- This means popular ecosystem artists accumulate real stream counts over the simulation — the world has a functioning music economy, not just the player.

Album Streams — Fixed Logic

- Album streams were previously double-counting. Fixed: **album streams = sum of all individual song streams**. There are no separate 'album stream' numbers — the album total is just the tracklist total.

Album Release Popularity — Revised

- Album release popularity bump reduced from **15** → **10**.
- Songs inside an album no longer generate individual release popularity bumps. The album as a whole gives the flat +10 — a 20-song album gets exactly +10, same as a 7-song album. The songs are part of the statement, not separate events.

Single → Album Stream Continuity

- When a single is included in an album, its stream history carries over intact. The song's stream count is identical whether viewed as a standalone single or as a track within the album.
- Other album tracks start from 0 streams at release and accumulate from that week forward.
- The single always appears **at any position** in the tracklist, not forced to track 1.
- Review score for the single stays **consistent** — the score doesn't change based on whether it's viewed as a single or an album track.

UI & Catalog Fixes

Issue	Fix
Multiple ft. artists only showing 1 in title	All collaborating lyricists/vocalists now appear comma-separated in the ft. credit
Songs still showing in Singles after being added to album	Once added to an album, a song is removed from the singles list — it lives in the album only
Metadata only viewable for album songs	Singles now also have a viewable metadata screen
No metadata in catalog manager for own songs	Catalog manager now includes a View Metadata option for all player songs

Cannot add features after adding song to album	Feature requests remain available for album tracks — the album tracklist is not locked
Cannot view album tracklist	Album view now shows full tracklist with song titles, features, quality, and streams
Growing artists excluded from player collaborations	Player can now collaborate with growing artists

View Artists — Enhancements

- Stream totals now visible in the View Artists section — each song listed with its total accumulated streams, for both singles and album tracks, in a dedicated column.
- New section per artist: **Top 5 Most Streamed Songs** — shows song title (with ft. credits), total streams, and release date. Updated every week as streams accumulate.

Rank	Title	Streams	Released
1	arose ft. kendrick lamar	142,300,000	Year 1, Wk 14
2	cold circuit	88,400,000	Year 1, Wk 8
3	drugs ft. drake, travis scott	61,200,000	Year 2, Wk 3
4	lost meridian	44,700,000	Year 1, Wk 22
5	pavement gospels	31,000,000	Year 2, Wk 11

→ *Top 5 reflects cumulative streams — a song released 40 weeks ago with steady numbers can still beat a recent hit with a short spike.*